

Cloud Engineer Internship Plan

▼ Expected Outcomes

By the end of this four-week Cloud Engineering internship program, interns will have developed and demonstrated proficiency in the following areas:

Docker & Containerization

- **Containerizing Applications:** Ability to package applications into Docker containers, including writing efficient Dockerfiles (using multi-stage builds and .dockerignore) and running/debugging containers locally.
- **Image Management:** Managing container images by tagging and versioning builds, and pushing/pulling images to a registry (e.g., Google Artifact Registry). This includes maintaining versioned Docker images and understanding layer caching for optimization.

GCP Infrastructure

- **Cloud Project Setup:** Setting up a Google Cloud project with proper configurations, including enabling APIs and creating IAM entities (users, service accounts, roles) necessary for deployments.
- **Resource Provisioning:** Creating and managing core cloud resources such as virtual networks (VPCs), virtual machines, and serverless services. For example, deploying a Compute Engine VM in a custom VPC with appropriate subnets and firewall rules, and deploying containerized applications to Cloud Run successfully.
- **Deployment and Troubleshooting:** Deploying applications on Google Cloud infrastructure (VMs or Cloud Run) and handling any issues that arise during deployment. Interns will be adept at diagnosing common problems with authentication, permissions, or networking and applying fixes to achieve a working deployment.

CI/CD Automation with GitHub Actions

- **Continuous Integration Pipelines:** Building automated CI pipelines using GitHub Actions to compile code, build Docker images, and run tests on each commit. Interns will configure triggers, manage secrets and caching, and document the pipeline logic.
- **Continuous Deployment Workflows:** Implementing CD workflows that authenticate to GCP and automatically deploy updated container images to Cloud Run. This includes managing credentials (service account keys, etc.), handling permissions for GitHub runners, and setting up a build → push → deploy sequence.
- **Pipeline Debugging and Optimization:** Learning to debug CI/CD failures (syntax errors, permission issues, etc.) and optimize pipeline performance (e.g., improving build times, implementing rollback strategies). Interns will ensure the deployment process is reliable and well-documented, including steps for rollback and recovery.

Infrastructure as Code with sst.dev

- **IaC Project Structure:** Initializing and organizing an Infrastructure-as-Code project using SST.dev, following best practices for project structure and stage configuration. Interns will define multiple environments (development, staging, production) and document the environment strategy.
- **Automated Resource Provisioning:** Mapping SST constructs to Google Cloud services (such as defining Cloud Run services, networks, and other resources in code) and deploying infrastructure programmatically. The goal is to have all cloud infrastructure provisioned through code with no manual steps, resulting in reproducible environments.
- **Maintainable Infrastructure:** Ensuring that all infrastructure configurations are version-controlled and maintainable. Interns will learn to update infrastructure via code, debug deployment issues in the IaC framework, and produce architecture documentation for the systems they build.

Security & IAM

- **Least-Privilege Access:** Designing and implementing IAM roles according to the principle of least privilege for all cloud resources (users, service accounts, and services). This includes assigning only necessary permissions and using separate service accounts for distinct functions (e.g., CI/CD pipelines vs. application runtime).
- **Permission Auditing & Troubleshooting:** Being able to troubleshoot and resolve authorization issues by diagnosing IAM permission errors and misconfigurations. Interns will maintain an **IAM access matrix** or similar documentation to audit who has access to what, and adjust roles as needed.
- **Secure Service Access & Networking:** Configuring secure access to services and networks. For example, making a Cloud Run service private (accessible only with proper IAM permissions) and setting up network controls like firewall rules to restrict ingress/egress traffic to allowed sources. Interns will understand how to protect resources using VPCs, subnetworks, and firewall policies while still enabling necessary connectivity.

Observability (Monitoring & Logging)

- **Centralized Logging:** Implementing application and infrastructure logging using Google Cloud's logging services. Interns will configure logs for Cloud Run and VMs, and demonstrate the ability to query and analyze these logs to debug issues. They will also learn to improve application log outputs for better traceability.
- **Monitoring & Alerting:** Setting up monitoring dashboards for key metrics (using Google Cloud Monitoring) and defining alerting policies for critical events. Interns will tune alert thresholds to reduce noise and ensure that alerts are actionable, creating a more reliable monitoring system.
- **Incident Response Readiness:** Developing runbooks and procedures for operational response. Interns will create documentation for handling alerts and perform failure testing (e.g., simulating service crashes or bad deployments) to practice incident response. They will also write postmortem reports to document issues faced and how they were

resolved, demonstrating an understanding of production readiness principles.

Standard Session Structure

- Concepts & guided walkthrough
- Hands-on lab
- Troubleshooting & review
- Documentation & submission

Daily deliverable: GitHub commit + README update

Recommended Resources

- Google Cloud Engineer <https://www.skills.google/paths/11>
- DevOps with Docker <https://courses.mooc.fi/org/uh-cs/courses/devops-with-docker>

Week 1 Docker + GCP Foundations

Day 1: GCP Setup & IAM Basics

- GCP project, regions, zones, APIs overview
- IAM fundamentals: users, service accounts, roles
- Create budget and budget alerts
- GCP CLI setup and authentication
- Create project, enable APIs, create deployment service account
- Troubleshoot auth, quota, and permission issues
- Document setup steps and common errors

Output:

Working GCP project with deploy-ready service account

Day 2: Docker Fundamentals

- Image vs container, layers, caching
- Dockerfile structure and `.dockerignore`
- Dockerize a simple API (Node/Python)
- Run and debug container locally
- Document how to run the app and common pitfalls

Output:

Application running locally in Docker

Day 3: Advanced Docker & Artifact Registry

- Multi-stage builds and image optimization
- Image tagging and versioning strategy
- Push and pull images from Artifact Registry
- Debug registry authentication issues
- Document image versioning and release flow

Output:

Versioned Docker images stored in Artifact Registry

Day 4: Deploying Containers to Cloud Run

- Cloud Run execution model, revisions, scaling
- Environment variables and basic secrets handling
- Deploy container from Artifact Registry to Cloud Run
- Debug runtime and permission issues
- Document deployment steps and service URL

Output:

Public Cloud Run service deployed successfully

Day 5: Networking Basics & Compute Engine

- VPC, subnets, routes, firewall rules
- Cloud Run networking vs VM networking
- Create custom VPC and deploy Compute Engine VM
- Configure SSH and firewall rules
- Document network diagram and access rules

Output:

VM running in custom VPC with correct firewall access

Week 2 GitHub Actions CI/CD + sst.dev IaC

Day 6: GitHub Actions Fundamentals (CI)

- CI/CD concepts and pipeline flow
- GitHub Actions triggers, runners, secrets, caching
- Build Docker image and run tests on push
- Debug workflow syntax and runner issues
- Document pipeline logic

Output:

CI pipeline building and testing Docker images

Day 7: Continuous Deployment with GitHub Actions

- Authenticating GitHub Actions to GCP
- Deploying to Cloud Run via CLI
- Build → push → deploy pipeline
- Debug permissions, registry access, and deploy failures
- Document rollback and deployment flow

Output:

Automated deployment to Cloud Run via GitHub Actions

Day 8: sst.dev Core Concepts

- Infrastructure as Code principles
- SST project structure and stages
- Initialize SST project
- Define environments (dev/staging/prod)
- Document environment strategy

Output:

SST project initialized and structured

Day 9: sst.dev + Cloud Run Infrastructure

- Mapping SST constructs to GCP services
- Define Cloud Run service using SST
- Deploy infrastructure via SST
- Debug deployments and outputs
- Document architecture and SST outputs

Output:

Infrastructure reproducible via SST (no console click-ops)

Day 10: Evaluation Project

Individual Evaluation Scope

- Dockerized application
- Infrastructure defined using SST
- GitHub Actions pipeline for build and deploy
- Image stored in Artifact Registry

Submission

- GitHub repository
- Cloud Run service URL
- README: architecture, CI/CD flow, infra design
- Short postmortem: issue faced and resolution

Week 3 Security, IAM & Networking in Practice

Day 11: Advanced IAM

- Least-privilege role design
- Service accounts for CI vs runtime
- Restrict CI permissions
- Debug IAM permission errors
- Document IAM access matrix

Day 12: Private Services & Access Control

- Public vs private Cloud Run services
- IAM-based service invocation
- Configure private access
- Debug auth and access errors
- Document access flow

Day 13: Networking in Practice

- Firewall rule patterns
- Ingress and egress control
- Test allowed and blocked traffic

- Network debugging techniques
- Document firewall rationale

Day 14: System Design for Cloud Engineers

- Stateless vs stateful services
- Scaling and failure patterns
- Design final capstone architecture
- Peer review and critique
- Document final system design

Day 15: Refactor & Hardening Day

- Improve Dockerfile quality
- Optimize CI pipeline performance
- Harden IAM roles
- Clean SST structure and stages

Week 4 Observability & Final Capstone

Day 16: Cloud Logging

- Logging concepts and structure
- Query logs for Cloud Run and VMs
- Improve application logging
- Debug issues using logs
- Document useful log queries

Day 17: Cloud Monitoring & Alerting

- Metrics and dashboards

- Create monitoring dashboards
- Configure alerting policies
- Tune alerts to reduce noise
- Document alert runbooks

Day 18: Production Readiness & Failure Testing

- Production readiness checklist
- Simulate failures (bad deploy, crash, permission error)
- Incident response workflow
- Write postmortem documentation

Day 19: Final Capstone Build

- Build end-to-end system
- SST-managed infrastructure
- Dockerized services
- GitHub Actions CI/CD
- Monitoring, logging, and alerts

Day 20: Final Evaluation

Capstone Requirements

- All infrastructure defined with SST
- Clean Dockerfile and reproducible builds
- GitHub Actions CI/CD pipeline
- Secure IAM with least privilege
- Monitoring dashboards and alerting
- Log queries and runbooks
- Live debugging demonstration

Submission

- GitHub repository
- Architecture, pipeline, IAM, and ops documentation
- Evidence: dashboards, alerts, logs